PeerEval: Learning Whose AI Judgments to Trust

Xinya Du, The University of Texas at Dallas

Arena Academic Partnerships Program, Fall 2026 12 months $50,000

Research objective

AI judges compare answers and decide which is better, but they can favor their own model family. Arena found a 37-percentage-point gap between three same-provider judges' preferences for their family and human votes [3]. **We propose to use mutual agreement to determine how much influence each AI judge receives.** A panel repeatedly scores each judge's agreement with other model families and updates its weight. The scientific question is when this feedback loop identifies useful judges and when it amplifies a shared bias.

Proposed method and preliminary evidence

Several language models review the same anonymous answer pairs. We start with equal judge weights and combine their votes into collective preferences. Each judge then receives an agreement score based on how consistently its votes match the weighted votes of judges from other model families. We smooth and normalize these scores to obtain the next weights, and repeat the calculation until weights stabilize or reach an iteration limit. The final weights determine answer preferences and model rankings. Judge influence is learned from peer reviews, with human votes held out to evaluate the results.

Our prior peer-evaluation work [1] and PeerEval leaderboard and tool [2] provide the infrastructure for the project. A pilot covers 1,179 human-labeled battles with four judges. On 794 crowd battles, agreement weighting matched 67.8% of votes, versus 65.7% for equal weighting. These exploratory results motivate testing when agreement is useful evidence and when it repeats a shared bias.

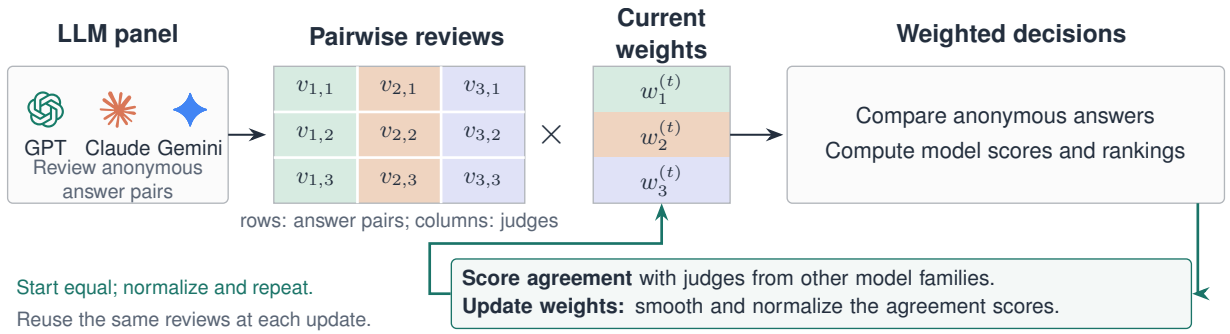


Figure 1: Proposed mutual-agreement weighting. Judges compare anonymous answers. Their agreement with judges from other model families determines the next weights, which are used to recompute collective preferences and rankings. Each iteration reuses the collected reviews. Icons illustrate model families.

Project scope and requested support. Months 1–3 will establish data and baselines; months 4–8 will develop the agreement loop; months 9–12 will complete transfer tests and release a paper, code, and evaluation reports. The $50,000 gift allocates $36,000 to graduate research, including fringe and tuition; $10,000 to APIs and computing; and $4,000 to evaluation annotations. At maximum scale, 10,000 pairs $\times$ eight judges $\times$ two orders require 160,000 judging calls; pilot costs will set the affordable scale. Results will complement Arena's voting and AutoEval [4].

Research approach

1. Update judge weights from mutual agreement. Six to eight judges from different families will review anonymous answer pairs in both display orders, choosing A, B, tie, or both bad. We will align the orders and average each judge's votes. At iteration t, judge i receives an agreement score $a_i^{(t)}$: its mean agreement with the weighted votes of judges *outside its own model family*. This comparison excludes the judge's own vote and family. Starting from equal weights, we update

$$w_i^{(t+1)} = (1-\eta)w_i^{(t)} + \eta \frac{\varepsilon + a_i^{(t)}}{\sum_j (\varepsilon + a_j^{(t)})},$$

where $0 < \eta \leq 1$ controls how quickly weights change and $\varepsilon > 0$ keeps every judge eligible. We stop when weights stabilize or reach a fixed iteration limit. The final weighted votes determine answer preferences and model rankings. No human labels enter these updates.

2. Determine when the weighting loop is reliable. Agreement can reward a shared mistake, so we will study what prevents a group of similar judges from reinforcing itself. We will compare ordinary agreement, agreement excluding a judge's own family, and a version that limits any one family's total influence. We will analyze convergence, sensitivity to initialization, and how many correlated judges it takes to change a decision. Peer-assessed answer-quality weights [1] and a fixed combination of answer quality and agreement will serve as comparisons. These tests separate the benefit of the update rule from simply adding more judges.

3. Test whether peer weights transfer to new questions and models. We will construct weights on development prompts, freeze them for unseen prompts, and compare this with recomputing weights on new unlabeled prompts. Later time windows and new answer-model families will test transfer. We will bootstrap questions and remove one judge family at a time to measure how stable the resulting rankings are. Instability will flag a result for additional review; we will test whether this flag predicts disagreement with people. For the proposed method, human votes are used only for evaluation.

Evaluation, outputs, and feasibility

We will use up to 10,000 public and, subject to agreement, recent Arena battles. Baselines include the strongest single judge, equal weighting, answer-quality weighting, static agreement weights, and weights fitted to limited human votes. We will measure preference accuracy, log loss with fixed probability smoothing, ranking agreement, stability, and cost, with paired bootstrap intervals that rerun the weighting loop. Length-matched tests, balanced longer/shorter-answer cases, and a length heuristic will check whether apparent gains come from verbosity. Repeated human votes and display-order metadata will help interpret disagreements. Shared biases may still limit gains; we will report those failure cases.

References

[1] Li R, Patel T, Du X. PRD: Peer Rank and Discussion Improve Large Language Model based Evaluations. TMLR, 2024. Paper.

[2] PeerEval. https://peereval-ai.github.io/; pilot results. 2026.

[3] Arena Team. Self-Preference of AI Judges. September 30, 2026. Article.

[4] Arena Team. Introducing AutoEval to the Arena leaderboards. July 30, 2026. Article.

Application form answers

Applicant: Xinya Du; Assistant Professor of Computer Science; The University of Texas at Dallas; Richardson, Texas; Department of Computer Science.
Email: xinya.du@utdallas.edu. **Website:** https://xinyadu.github.io/.
Publications: https://xinyadu.github.io/publications.html.
Project title: PeerEval: Learning Whose AI Judgments to Trust.
Duration / funding: 12 months from receipt of funding; $50,000.

3–5 sentence summary of proposed research
We propose to use mutual agreement to determine how much influence each AI judge receives. A panel compares anonymous answer pairs, scores each judge's agreement with other model families, and repeatedly updates normalized weights. The final weights produce collective preferences and model rankings through our PeerEval platform (https://peereval-ai.github.io/). We will analyze convergence and shared biases, then test accuracy and transfer on Arena data with human votes held out for evaluation. Outputs will include a paper, open-source code, and reproducible evaluation reports.

Additional dataset access and why it is needed
We request up to 10,000 recent, de-identified text battles with prompts, paired responses, model identifiers, preference outcomes including ties and both-bad, coarse timestamps, and per-vote display order. Randomization metadata and repeated independent votes on the same response pairs in both orders, where available, would help separate order effects from preference ambiguity. Recent battles enable tests on new models and changing preferences. Public data support the core study; additional access would follow Arena's agreements.

Expected outputs (paper, dataset, code release, etc.)
A paper on mutual-agreement weighting, its assumptions, and failure cases; an open-source implementation in PeerEval; and reproducible evaluations of preference prediction, model rankings, transfer, stability, and cost. We will release permitted judgments and analysis code, with restricted data governed by the access agreement. We welcome a joint blog post or talk with Arena.

Brief budget explanation (3–5 sentences)
Approximately $36,000 will support graduate research, including applicable fringe and tuition. Model APIs and computing will receive $10,000, with pilot costs determining the final panel and sample size. Independent evaluation annotations will receive $4,000. Allocations will follow UTD requirements; no indirect costs are included unless required by institutional policy.

Why is this work relevant to Arena?
Arena's self-preference findings show why a panel needs more than a simple vote, while its human preferences provide a strong independent test of AI judges. Our project will determine whether models can learn useful evaluation weights from their own peer reviews and when that process reinforces bias. The results could complement AutoEval, support evaluation of newly released models, and identify unstable rankings that warrant further review.

Requested resources and proposal link
Arena datasets; collaboration with Arena researchers. Joint publicity: Yes. Proposal: two-page submission PDF on GitHub.